Session 6: Two-Population Inference & t-Tests

Context & Objective

Textbook: Chapter 10 -- Hypothesis Testing for Two Populations

Objective: Students run a two-population t-test to compare conversion performance between two website layouts, testing for a significant difference in means.

1. Antigravity Skill Definition (two-population-t-tester)

Skill Name: two-population-t-tester

Purpose: Performs two-sample independent t-tests. Computes standard error, degrees of freedom, t-statistic, and p-value. Outputs a reject-or-fail-to-reject decision a manager can act on.

1.1 Core Logic & Formulas

Step 1: Gather the Inputs

Sample size (n1, n2), means (mean1, mean2), standard deviations (s1, s2)

Step 2: Compute Standard Error of the Difference

$s_diff = sqrt(s1^2/n1 + s2^2/n2)$

Step 3: Calculate t-Statistic

$t = (mean1 - mean2) / s_diff$

Step 4: Degrees of Freedom (Welch-Satterthwaite Approximation)

df accounts for unequal variances. Determines the critical value from the t-distribution.

Step 5: Compute p-Value and Apply Decision Rule

If p-value ≤ α: Reject H0. If p-value > α: Fail to Reject H0.

2. Student Execution Prompt

Evaluate if Layout A leads to higher daily sign-ups than Layout B.

2.1 Given Data & Test Setup:

Test Type	One-tailed (upper-tailed)
H0: Null Hypothesis	Layout A and B have equal mean daily sign-ups
Ha: Alternative Hypothesis	Layout A mean > Layout B mean
Sample sizes	n1 = 15 days, n2 = 15 days
Means	Layout A: 150 sign-ups/day, Layout B: 140 sign-ups/day
Standard deviations	s1 = 12.0, s2 = 15.0
Significance level (α)	0.05 (5%)

2.2 Assignment Task:

1. Calculate standard error of the difference.
2. Compute the t-statistic.
3. Determine degrees of freedom.
4. Find the critical value and p-value.
5. Make a reject or fail-to-reject decision.

3. Output & Interpretation

3.1 The Numbers:

Standard Error	4.966
t-Statistic	2.013
Degrees of Freedom	26
Critical Value (t_0.05)	1.706
p-value (one-tailed)	0.027 (2.7%)

3.2 Decision Rule: Critical Value Method

t-statistic (2.013) > t_critical (1.706) → Reject H0

3.3 Decision Rule: p-Value Method

p-value (0.027) < α (0.05) → Reject H0

3.4 Decision: Both Methods Agree. Reject H0

3.5 What It Means:

If Layout A and Layout B truly pulled the same sign-ups, the chance of seeing a gap this large is only 2.7%. That is low enough to reject the idea that they are equal. The data point toward Layout A outperforming Layout B.

This is not borderline. The t-statistic sits clearly past the cutoff, and the p-value is less than half of alpha. The difference is real.

3.6 What the Manager Does:

Deploy Layout A permanently. The test proves the higher sign-up rate is not random. Over 30 days, this 10-signup daily edge adds up to 300 more users. Confidence level is 97.3% (1 minus p-value). Risk of a false positive is low. Proceed with rollout.

4. Inference

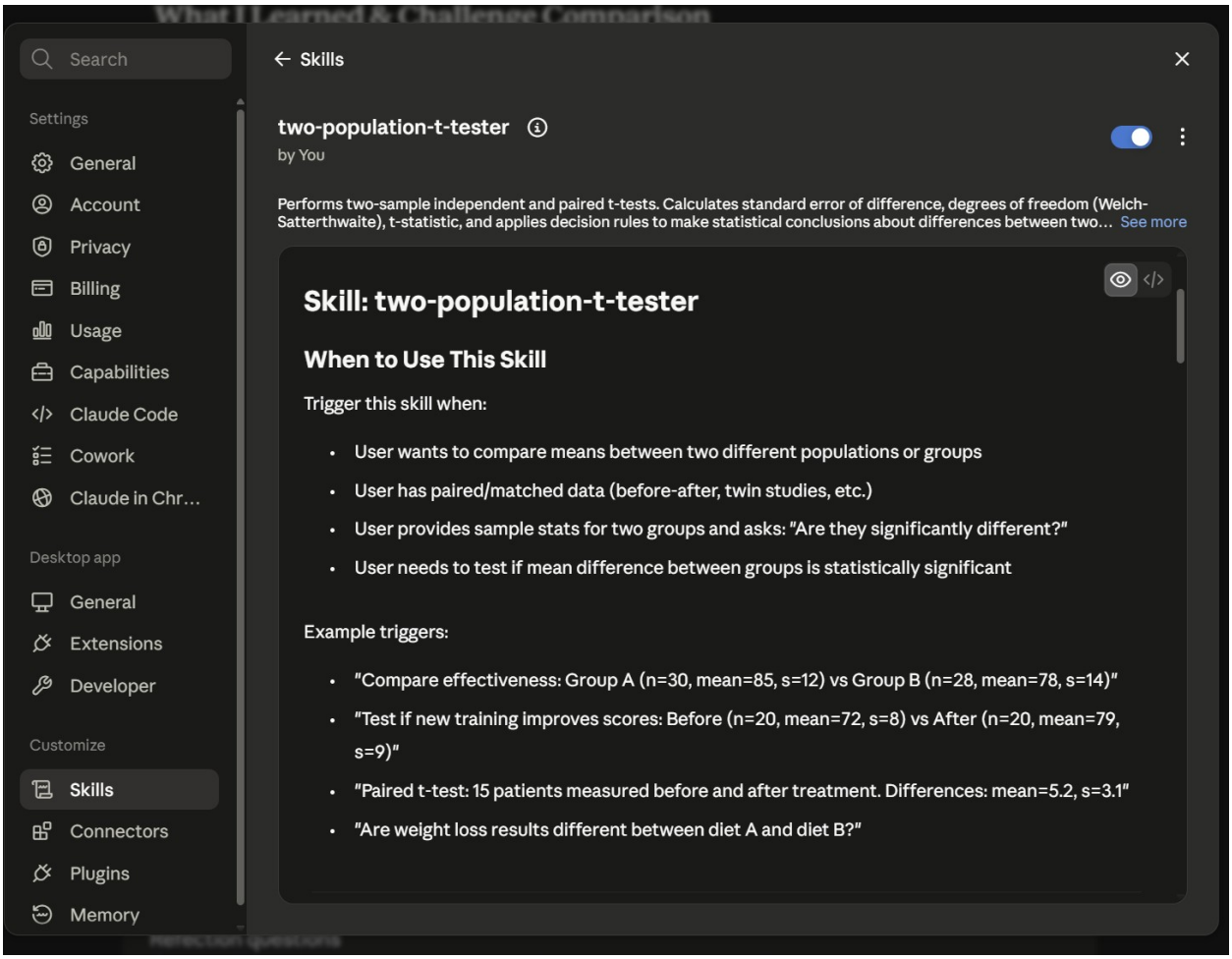
The critical-value method and the p-value method give the same answer every time. They are two sides of the same coin.

A p-value is the chance of getting a result this far from H0 if H0 is actually true. It is not the chance that H0 is true.

When $p \leq \alpha$, reject H0. When $p > \alpha$, fail to reject H0. Do not accept H0—just acknowledge a lack of evidence against it.

Small p-value plus large practical difference equals a strong business case. Small p-value alone is not enough.

Misinterpreting p-values is common. A small p means your data are unlikely if H0 is true, not that your effect is important. Always pair statistics with business judgment.


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Skills
two-population-t-tester
by You
Performs two-sample independent and paired t-tests. Calculates standard error of difference, degrees of freedom (Welch-Satterthwaite), t-statistic, and applies decision rules to make statistical conclusions about differences between two… See more
Skill: two-population-t-tester
When to Use This Skill
Trigger this skill when:
User wants to compare means between two different populations or groups
User has paired/matched data (before-after, twin studies, etc.)
User provides sample stats for two groups and asks: "Are they significantly different?"
User needs to test if mean difference between groups is statistically significant
Example triggers:
"Compare effectiveness: Group A (n=30, mean=85, s=12) vs Group B (n=28, mean=78, s=14)"
"Test if new training improves scores: Before (n=20, mean=72, s=8) vs After (n=20, mean=79, s=9)"
"Paired t-test: 15 patients measured before and after treatment. Differences: mean=5.2, s=3.1"
"Are weight loss results different between diet A and diet B?"